

Charles Stephen Thompson

AI Automation & Integration Engineer · Platform / Infrastructure · IAM & SSO

✉ hello@cst.nyc

☎ +1.332.262.8079

📍 Brooklyn, NY 11238

🌐 mrcharlesthen

🌐 cst.nyc

Automation and integration engineer with 14+ years connecting systems that were never designed to talk to each other — identity, CI/CD, monitoring, and now LLM agents. I design the interfaces and guardrails that let automation act on production safely: custom MCP (Model Context Protocol) servers exposing git/CI, metrics, DNS, and SIEM to AI agents under least-privilege scopes; n8n workflows that triage and remediate rather than just notify. Backed by a decade of enterprise IAM/SSO, platform engineering, and running a 40+ service private cloud where the automation has real consequences.

EDUCATION

Bachelor of Arts, Modern American Legal History
The University of Chicago — Chicago, IL

2005 – 2010

BA Thesis on the misuse and abuse of copyright law as compared to that seen in the patent field.

SKILLS

AI & AUTOMATION

Claude / LLM APIs

MCP Servers

n8n

Ollama

RAG

Prompt Engineering

HOME & BUILDING AUTOMATION

Home Assistant

Homebridge / HomeKit

Matter

Zigbee

Z-Wave

IDENTITY & ACCESS

IAM

SAML / OIDC / OAuth2 (PKCE)

Okta

Active Directory

PROFESSIONAL EXPERIENCE

Principal Automation & Platform Engineer

Jan 2022 – Present

BKHD Engineering — Brooklyn, NY

- Architected an AI automation platform where n8n workflows orchestrate LLM agents (Claude) to triage monitoring alerts, reconcile DNS, and drive CI/CD across a 40+ service private cloud — closing the loop from detection to remediation rather than stopping at notification
- Designed and shipped custom MCP (Model Context Protocol) servers exposing Forgejo git/CI, Grafana metrics, Technitium DNS, and a Wazuh SIEM to AI agents under per-scope least-privilege authorization, addressing the unsolved problem of letting agents act on production safely
- Built and deployed a Go/Vue log-analytics service end to end — TimescaleDB schema, syslog ingestion, and OIDC/PKCE authentication — replacing a Python predecessor under sustained production load
- Engineered a rootless Podman container platform (systemd quadlets) with multi-architecture (arm64/amd64) CI via Forgejo Actions and automated dependency management
- Architected DNSSEC-signed, split-horizon DNS with catalog-based zone replication, and high-availability PostgreSQL (Patroni/TimescaleDB) with automated failover
- Instrumented full-stack observability (Grafana, Prometheus, VictoriaLogs, InfluxDB) and a Wazuh SIEM; root-caused and remediated live outages spanning DNS blackholes, container-runtime deadlocks, and memory-pressure cascades
- Contribute upstream to open-source infrastructure projects and publish GPG-signed Debian packages and npm modules through a self-hosted registry with multi-arch build pipelines

Kerberos/LDAP

ADFS

Pocket ID

GSuite

AUTOMATION & CONFIGURATION

Ansible

Chef

CloudFormation

Jenkins

MDM

Terraform

CircleCI

GitHub Actions

Forgejo Actions

Renovate

LANGUAGES & SCRIPTING

Bash

CoffeeScript

Go

Groovy

HTML

JavaScript

jQuery

Node.js

PHP

Python

Ruby

Shell Scripting

SQL

Swift

TeX

TypeScript

TOML / XML / YAML

Vue

PLATFORMS & MONITORING

Active Directory

AWS

Azure

Cloudflare

Docker

Elasticsearch

GCP

GitHub

Grafana

InfluxDB

JAMF

JumpCloud

Kibana

Kubernetes

Logstash

Nagios

Okta

Prometheus

Splunk

UniFi 802.1X

Zabbix

Podman

Technitium DNS / DNSSEC

Pocket ID

VictoriaLogs

Wazuh (SIEM)

TimescaleDB

TECH STANDARDS & CONCEPTS

Agile Workflow

CI/CD

Cloud Computing

Directory Services

Distributed Computing

DNS

IAM

LAMP Stack

Microservices

Monitoring

Remote Config & Management

Founder & Principal Consultant

Jan 2018 – Present

Brooklyn Helpdesk, Inc. — Brooklyn, NY

- Deliver IT and DevOps engineering across ~500 managed client endpoints: IAM/SSO integration, CI/CD automation, and LAN/WiFi design, installation, and maintenance
- Standardized client identity on single sign-on (Okta, GSuite, Active Directory) with automated device lifecycle management via MDM
- Deliver home and office automation with Home Assistant and Homebridge/HomeKit — presence-aware lighting, climate, window coverings, and A/V scene control over Zigbee, Z-Wave, and Matter

Director of IT & Infrastructure Engineer

Nov 2019 – Dec 2021

Symbiont.io — New York, NY

- Reduced onboarding time by 25% and costs by 12% via streamlined IAM/MDM integrations (Active Directory, GSuite, Okta, JAMF) to achieve SSO from hardware to cloud
- Designed and deployed automation software integrations for corporate IAM, user management, and hardware de/provisioning
- Deployed, monitored, and maintained Kubernetes and Docker stacks for a team of 36 engineers

Software Infrastructure Engineer & Interim Director of IT

Blink Health — New York, NY

Aug 2018 – Feb 2019

- Provisioned all Windows server instances with focus on Active Directory (fault tolerance, high availability) using Chef, Ansible, and Terraform
- Automated AD credential writing to GSuite, reducing onboarding steps from 40 to 4
- Leveraged Google SAML to reduce external attack surface and lower cost of SSO delivery
- Integrated Windows/Mac/Linux/GSuite auth via LDAP providing cross-platform SSO for VPNs, databases, and critical applications

Lead IT Infrastructure Engineer

Oct 2016 – Aug 2018

Blink Health — New York, NY

- Designed and managed UniFi 802.1X LAN/WiFi infrastructure at three sites with AWS connectivity via site-to-site VPN for 200+ staff
- Achieved infrastructure-as-code with GitHub PR support, CircleCI integration, and Terraform revision history across multiple AWS accounts
- Managed annual technology budget of \$5mm+ with inventory comprising hundreds of endpoints
- Grew IT team to 10 members across three geographic locations; oversaw opening of four offices in three states

SaaS

SAML / OIDC

Version Control

MCP (Model Context Protocol)

AVP, Technology – Level III System & Network Engineer

Build America Mutual Assurance Company — New York, NY

Feb 2013 – May 2016

- Designed and implemented Active Directory backbone for SSO across Windows, Mac, and Linux using AD, Kerberos, LDAP, and MFA
- Implemented Jabber/XMPP with SSO and VOIP integration for the pre-existing phone system
- Migrated servers and applications to cloud providers, achieving \$250K+ in savings over five years
- Conducted Sarbanes-Oxley compliance reviews; supported nearly 100 users across three locations

Support Technician II

Jan 2012 – Jan 2013

SecondMarket, Inc. — New York, NY

- Configured SAML/ADFS SSO for Yammer, Salesforce, and Zendesk — saving the company \$36 per user per year
- Managed \$1.5M+ annual technology budget across three locations (New York, San Francisco, Hong Kong)